

Plant Disease Detection Using Quantum Image Processing

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Abstract—EDQRCI allows automation in order to create, establish quantum images and it converts color pixel in well-defined ways. Our model is independent of image rectification process and algorithm, although storage and other necessary process. EDQRCI generally make use of the same idea of energy metamorphose, related to 2D mapping in order to encode values of energy to group of quantum data. Suppose, if the data given is electromagnetic, then it will be processed and stored, then the data given can be encrypted to quantum mode in order to represent color data. On the other hand, if the given data is infrared radiation, then the encoding of quantum data into color representation can be done based on the intensity of radiation. Different Quantum Image Description permits the portrayal, development and acquisition of data in individual modes in quantum mechanism for another experimental implementation.

Index Terms—QIP (Quantum Image Process), EDQRCI (Enhanced Double quantum images encryption), Quantum Computing.

I. INTRODUCTION

Most prompted field in quantum information and computation is quantum image processing. Quantum Computation is the sector that finds the estimation power and other properties of computer based on quantum mechanical theory. This part initiated in the early 1980s with propositions for analog quantum computers and hold out more digital ground, when in 1985 David Deutsch explained the universal quantum turning machine. Quantum computing plays an important role in advanced computing but details taken out in which quantum computational operations were taken out in a very minuscule number of quantum bits. In many of the government organizations & companies are supporting to fund quantum computing research in order to establish quantum computers for medical screening, business, civilian, research & education and also in security purposes. The representations provide the essential proof that quantum image processing can reduce the processing speed and secure transmission. computational complexity. The quantum analogous of the classical bit is the qubit. In quantum systems qubit can be described as the super position of two or the normal states. A quantum registers is denoted by a

sequence of qubits. if qubit quantum register is represented as 's' and operator in the states space H is represented as 'U', then registers applied to the operator U is called a quantum gate. In the recent year Pushpa B R et al [1], introduced a design for the detection and identification of crop leaf disease using CNN based AlexNet Model and their paper has achieved an accuracy of 96.76%. [2] This paper Z Saeed et al, describes classification method based on DCNN such as ResNet-152 and InceptionV3 for potato disease and has acquired an accuracy of 98.34% and 95.24 %. Eisha Akanksha et al [3] proposed a methodology named (OPNN) which is used for classification. This model has achieved maximum accuracy of 95.5 % which is similar to existing models. [4] proposed quantum image processing (QIP) to show real quantum superiority. Malik et al [5] proposed a model to ensure high quality image encryption scheme by improving bit plane by using Quantum chaotic behavior & RC4. B J Bipin Nair et al [6] proposed a work where in first stage to get threshold image Butterworth filter and otsu thresholding is performed. Next to that in second level to obtained masked image. Grape leaves is used for treating syndrome of chronic fatigue syndrome, bleeding, diarrhea, etc. Grape also controls the glucose level in our blood. The grape plant one of the most planted fruit around the world as they are widely used for making wine. Pests like -Japanese Beetles (Popillia japonica), Black Vine Weevil (Otiorynchus sulcatus), Grape Cane Girdler (Ampelogypter ater), Grape Mealybug (Pseudococcus maritimus), Glassy-Winged Sharpshooter (Homalodisca vitripennis) are the most commonly seen pests that cause numerous diseases in the grape leaf. R. Sujatha et al [7] In their work, all deep learning methods are analyzed & the outcomes of DL is remarkable compared to Machine learning. Sunil C. K [8] In this proposed method U^2 -Net is used to remove complex background and for classification CNN, EfficientNet, and EfficientNetV2. The models achieved an accuracy of 98.26%. Minu Eliz Pothen et al [9] proposed disease detection of infected plants can plan numerous management of disease. Diseased image will be segmented by applying Otsu's method and accomplished 94.6 %. Pushpa B R et al [10] designed an system to classify medical plants by employing differ-

ent texture features has been implemented in their research paper. Junde Chen et al [11] proposed VGGNet which can be changed by swapping the final layer with an CNN layer and therefore uses the Swish trigger function to replace ReLu directly. Reem Ibrahim Hasan et al [12] This review examines and discusses the latest methods developed in three years including how these methods can be used in classifiers and their implications on the classifier accuracy. Kawcher Ahmed et al [13] In this paper, the data-set were trained using variety of ML algorithms. Decision tree algorithm has achieved greater than 97 % accuracy. Shima Ramesh et al [14] In this paper, Random Forest helps in classifying both healthy and diseased images where the dataset are trained. Features from an image are extract using (HOG) histogram. The model has acquired an accuracy of 70%. K.S. Archana et al [15] proposed work have various hybrid image segmentation techniques and classification algorithms were analyzed and an automatic detection method was used to identify the specific diseases under different environmental conditions. R.P Narmadha et al [16] proposed a method to remove noise and errors made by humans and minimize time taken to menstruate the effect of rice leaf disease and it also increases accuracy. Aparajitha et al [17] proposed, the adaptive threshold is applied for segmenting the diseased part of the leaf image. The result was an accuracy of 96%. F. Yan et al [18] paper has reviewed current main stream quantum image representation and has discussed the advancement made in the area. Based on their paper most of the review was on the FRQI representation and its applications. Sachin D. Khirade [19] The proposed work has discussed the method for the detecting the diseases, segmenting the disease and feature extraction algorithm used. Anjali et al [20] designed an work on grape disease detection & classifies using SVM classification technique. And the system successfully detect and classified the studied disease with an accuracy of 88.89%. Simona Caraiman et al [21] Considered HOG method for image processing methods. And for development HOG calculation and threshold-based segmentation. Jayme Garcia ArnalBarbedo [22] presents overall methods using DIP for detection, and classification of data-set from digital images in the clear spectrum. P.Revathi et al [23] expressed technologies using moving symptoms from images of cotton plant and classified using HPCCDD algorithm. Phuc Q. Le et al [24] proposed FRQI method a representation of images in quantum computers. The results imply that the FRQI could play a crucial role in image rendering and processing in quantum computers. Dheeb Al Bashish [25] proposed image processing based approach has been proposed in this work. Segementation using k mean and the segmented datasets will be passed through pre-trained neural networks. They acquired and 93% accuracy.

II. QUANTUM IMAGE FORMAT

QIF is the origin of Quantum Information Processing. Qubit Lattice, Real Ket and FRQI are 3 primary quantum image formats. Venegas-Andraca proposed Qubit Lattice which is the first QIF. He commented that the constancy point of waves in light will be mapped into amplitude probability of

qubit, so the pixel values of i^{th} and j^{th} columns will be saved as shown in eq(1), and overall the image will be denoted in qubit string (eq(2)).

$$\begin{aligned} |pixel_{i,j}\rangle &= \cos \frac{\theta_{i,j}}{2} |0\rangle + \sin \frac{\theta_{i,j}}{2} |1\rangle, \\ |image\rangle &= \{|pixel_{i,j}\rangle\}^i \\ &= 1, 2, \dots, n_1, j=1, 2, \dots, n_2 \end{aligned}$$

This has been used to graph the dimensional information of the image into single qubit's amplitude without implementing quantum properties. By utilizing quantum superposition state, an advanced form of single qubit was formed earlier known as FRQI. Graph of every pixel gray scale measures to amplitude, simultaneously, introduce a auxiliary qubit to represent spatial location of every pixels. Overall image is ready into huge quantum superposition state. Due to the excellence of FRQI, most of research is carried to extended project. A set of pixel state values was used by Zhang at al to represent the state instead of portraying the pixel's gray value to the amplitude angle. This pattern ease different image operations and to improves efficiency. Later another researcher elongated FRQI to polar coordinates and retrieved pixel's geographical position with polar diameters and polar angles. Real ket, the previous QIF divides the image into a 2X2 pieces and then sum up the four pixels gray-scale values into the possible dimensions of every component of quantum state with 2 qubits. The standardized state of quantum is called qudit. Qudit is the superpositional states of two qubits.

III. PROPOSED METHOD

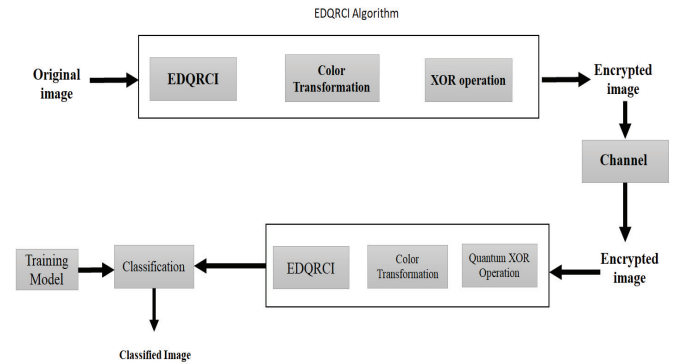


Fig. 1. Training Model

IV. QUANTUM IMAGE OPERATION

In Recent years, quantum domain became controversial subject for discussion. Classifying quantum operation to geometric transformation, color transformation, compression and retrieval operations.

A. Geometric Transformation

Geometric Transformation in general works on the position component of image operation operator it is also an a very basic operation in digital image processing. The operators are

classified to global and local operation. Which operates on all pixels of the image like in all angle of the image. The equation below, represents operation, where G represents the unitary transformation in each pixels of the images.

$$G(|image\rangle) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} |C_k\rangle \otimes G(|k\rangle).$$

As, the result global operation G's is considered as more efficient than a classical image.

B. Color Transformation

The transformation of image's pixel data, mostly processing of halftone images. These data will be retrieved by some factors like image encoder/decoder, picture watermark, etc. In case of factors for color transformation, these can be divided into global operation and local operation. The color transformation, gradually increases the acceleration in comparison to the classical counter path. Local operation performs good speed compared to classical counter path. As we know the quantum Fourier transform is fast compared to classical one, it is commonly used for watermarking and picture encoder and decoder. This mechanism performs information security for encoding and decoding of image.

C. Complicated Image Operations

For achieving compression, restoration, identification, segmentation, registration and related other operation in classical image processing. For converting the image data it has to perform some other pre processing method based on the original pixel data of the image. Such operations are known as "complicated image operation" which is used for image compression and image restoration.

D. Image Compression

The limiting of quantum gates while processing quantum data is known as image compression. This method is used to compress the count the number of quantum gates by solving expression (Boolean Expression). This mechanism is not similar to classical Compression of image.

E. Image Retrieval

This is an another important method frequently used in QIP. It has mainly two concepts: Firstly, from quantum state to restore classical data and others to extract the quantum data into quantum state. In order to find easy patterns in black-white binary data Schutzhold proposed a probabilistic quantum algorithm which uses quantum fourier transform method which achieved exponential speed comparing to classical algorithm.

V. DATASET

SI.NO	DISEASE	NO.OF DATASETS
1	Grape Esca	55
2	Grape leaf blight or Isariopsis Leaf Spot	55
3	Grape leaf blight (Black rot)	55
4	Healthy Grape leaf	55

VI. WHAT I WANT TO ADDRESS

Increases the speed of the operation when it is deployed in edge extraction, security and de-noising of images process, while using Color Transformation technique images can be restored and then the retrieved data can be encoded /decoded, image watermark and shuffling of color blocks can be achieved. By transforming the original image data, image compression, restoration, recognition, segmentation, registration and other operations can be done by making some pre-processing. This can also be used as function mapping in order to convert energy levels into an group of quantum states. These Quantum States can be detected and recorded if the energy levels are electromagnetic waves.

VII. RESEARCH GAP

This Technique can only be used in Quantum Image Processing though it is a unique operation. Thus any non-unique operations cannot be used in Quantum computing. The main theoretical problem in this process is the quantum noise, thus it has some quantum error, these error may lead to have some loss of quantum data. Progress of fields may not be achieved by expressing, in order to use this approach more image processing operations is required.

VIII. METHODOLOGY

In this research, we proposed a method to address detection of using quantum image processing this might help in identifying disease and will help in taking necessary precautions in the very early stage. So that the spreading of the disease to various other parts can be controlled. Dataset used in this work are collected from an online repository named kaggle with colored image. The work has been carried out in python programming and specifically we have taken images of grape plant (both diseased and healthy). The most common grapes disease are powdery mildew, anthracnose, greenaria bitter rot, black rot, alternaria, rhizopus rot, etc. For our study we have considered diseases such as alternaria blight, black rot and black measles. Color Transformation (Pre-Processing) and EDQRCI (Methodology inspired from DRQI) are used for more accurate and faster result. EDQRCI is an extended study of previously proposed DRQI model. DRQI, is a 4 channel model. But in our research we are reducing one channel and will give the same accuracy as the previous DRQI method.

Basically, there will be a training data set and their will be a input dataset. So the input will be set with some parameters (height, ...). So, that will be the input and the input image might be a color image. So we will convert that image into a gray scale. Next to that the converted image will be converted to an array. And we will plot the area, select the area next to that we will give to the method and will extract features. After extraction we will keep comparing extraction with the qubit gates and will be compared with the training data set. So, when this part and the existing part both are same based on that we will make the result.

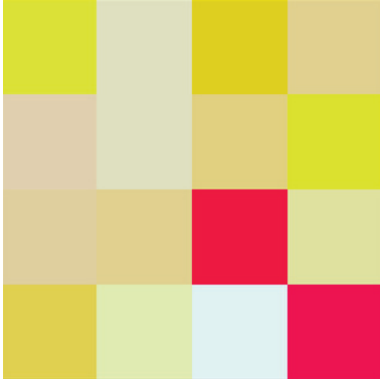


Fig. 2. Training Model

A. Enhanced Double quantum images encryption (EDQRCI) for Digital Images:

EDQRCI proposed with corresponding quantum circuit are looked into, in which 2 color images can be stored concurrently to a superposition state. Through our execution, EDQRCI representation has many advantages. It help in saving more of space because in color image of size $2n \times 2n$, employs only $2n+9$. But to encrypt same size of image we need $(2n + 24) \times 2$ qubits. It is different from FRQCI, because color information are saved as the probability amplitude of 3 single qubits and with low complexity image processing operations can be performed.

Input - Color image

```
Grape_Esca_(Black_Measles)
[>.....] - ETA:
111116/16 [=====]
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Fig. 3. Sample of Detected Disease

B. EDQRCI quantum image representation model and preparation:

Previous DQRCI representation model stores $2n \times 2n$ digital image where each channel range between $[0, 255]$. Color information of the channels (ie, RGB color) is encoded to 3 different entangled qubit sequence correspondingly.

$$|I\rangle = \frac{1}{\sqrt{2^{2n+3}}} \sum_{L=0}^{2^3-1} \sum_{Y=0}^{2^n-1} \sum_{X=0}^{2^n-1} |C_L(Y, X)\rangle \otimes |L\rangle \otimes |YX\rangle$$

Here $|L\rangle$, $|Y\rangle$ & $|X\rangle$ represent bit-plane & position info $|C_L(Y, X)\rangle$ denotes color info of (Y, X)

educated by DQRCI, and by making use of quantum superposition and entanglement, we have developed EDQRCI representation model with a channel ranged $[0, 255]$.

The EDQRCI expression is as,

C. Color complement operation

Color channel of every resultant image is the complement of the corresponding color channel of the real image. In the complement image, the leaves of the plant seem purple

$$|D\rangle = \frac{1}{\sqrt{2^5}} [|10000\rangle \otimes |000\rangle \otimes |00\rangle + |00001\rangle \otimes |000\rangle \otimes |01\rangle + |001000\rangle \otimes |000\rangle \otimes |10\rangle + |110000\rangle \otimes |000\rangle \otimes |11\rangle + |100000\rangle \otimes |001\rangle \otimes |00\rangle + |000101\rangle \otimes |001\rangle \otimes |01\rangle + |101000\rangle \otimes |001\rangle \otimes |10\rangle + |110000\rangle \otimes |001\rangle \otimes |11\rangle + |110010\rangle \otimes |010\rangle \otimes |00\rangle + |100101\rangle \otimes |010\rangle \otimes |01\rangle + |011000\rangle \otimes |010\rangle \otimes |10\rangle + |110111\rangle \otimes |010\rangle \otimes |11\rangle + |100000\rangle \otimes |011\rangle \otimes |00\rangle + |010001\rangle \otimes |011\rangle \otimes |01\rangle + |001110\rangle \otimes |011\rangle \otimes |10\rangle + |110000\rangle \otimes |011\rangle \otimes |11\rangle + |100000\rangle \otimes |100\rangle \otimes |00\rangle + |000101\rangle \otimes |100\rangle \otimes |01\rangle + |101000\rangle \otimes |100\rangle \otimes |10\rangle + |110000\rangle \otimes |100\rangle \otimes |11\rangle + |110010\rangle \otimes |101\rangle \otimes |00\rangle + |100001\rangle \otimes |101\rangle \otimes |01\rangle + |111000\rangle \otimes |101\rangle \otimes |10\rangle + |110111\rangle \otimes |101\rangle \otimes |11\rangle + |110010\rangle \otimes |110\rangle \otimes |00\rangle + |110001\rangle \otimes |110\rangle \otimes |01\rangle + |011110\rangle \otimes |110\rangle \otimes |10\rangle + |110111\rangle \otimes |110\rangle \otimes |11\rangle + |100000\rangle \otimes |111\rangle \otimes |00\rangle + |010101\rangle \otimes |111\rangle \otimes |01\rangle + |001110\rangle \otimes |111\rangle \otimes |10\rangle + |110000\rangle \otimes |111\rangle \otimes |11\rangle]$$

Fig. 4. EDQRCI expression

because the red & blue signals are greater than the green signal. And in the complement of a gray scale or color image, pixel value of each is subtracted from the max pixel value supported by the class. The differentiation used as the pixel value in the output image. Darkish region becomes lighter and light region becomes darkish in the outcome.

Color complement operation UC defines as $UC = X^{\otimes 6} \otimes I^{2n+3}$.

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Here x denotes quantum NOT gate.

D. Color transformation

$$U = U_R U_S^2 U_S^1 U_C$$

Here U is reversible, & $U_{-1} = U_C U_S^1 U_S^2 U_R$.

Took two colour images with the size of 512×512 which displays the outcomes only after the application of transformation U. So the proposed color transformation U will be able to alter 2 original input image into absurd one, visible delicacy will be lost. Transformation 'U' is used to encode the method into cache the original image development.

TABLE I
ACCURACY EVALUATION OF PROPOSED METHOD.

Samples	Proposed Method(%)
Sample 1	94.5
Sample 2	96.3
Sample 3	93.6
Sample 4	91.8
Sample 5	91.3
Sample 6	98.1
Sample 7	92.7
Sample 8	80.4

E. EDQRCI color image encryption

EDQRCI color image encryption consists of 2 stages. Initially, the quantum color transformation is executed and next to that color & info of bit-plane without replacing

the position info. To improve the security of the encoding scheme & to make pixel measures unchanging, quantum XOR operation will modify the color information. While information of the bit-plane & position will persist constant.

TABLE II
GRAPH REPRESENTING COMPARISON OF PERFORMANCE METRIC OF PROPOSED METHOD

	Accuracy	Precision	Recall	F score
Image 1	94.5	86.1	84.1	85.1
Image 2	96.3	84.9	84.9	84.9
Image 3	93.6	85.8	84.1	84.9
Image 4	91.8	85.3	85.3	85.3
Image 5	91.3	84.9	85.7	85.1
Image 6	98.1	84.4	86.1	85.2
Image 7	92.7	85.8	84.5	85.2
Image 8	80.4	85.6	85.3	85.4

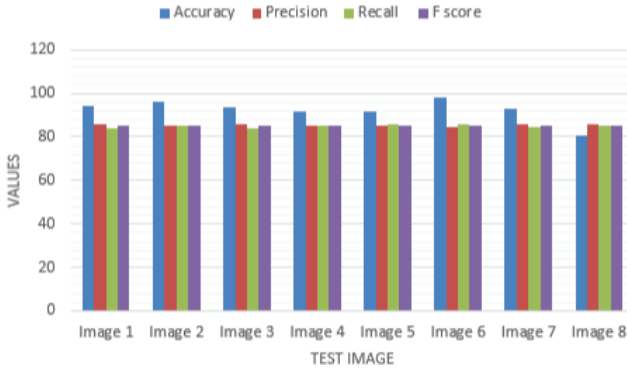


Fig. 5. Graph representing performance evaluation measures of proposed method

F. EDQRCI color image decryption scheme

Step 1: $(|T^1i\rangle, |V^1i\rangle, |W^1i\rangle, |T^2i\rangle, |V^2i\rangle, \text{and } |W^2i\rangle)$ can be generated with values 0. After the application of quantum XOR operation U_{XOR} called by $(|T^1i\rangle, |V^1i\rangle, |W^1i\rangle, |T^2i\rangle, |V^2i\rangle, \text{and } |W^2i\rangle)$ to $|E\rangle, |E\rangle$ can be got.

Step 2. quantum inverse color transformation will be applied upon E^1 & so we will get the decrypted image.

IX. CONCLUSION

EDQRCI, is a newly discovered and is an extended study of existing DQRCI method. In EDQRCI, the color information is placed in qubit sequence i.e., In the 1st qubit sequence 2 images are stored, while in the 2nd and 3rd qubit sequence the information of bit-plane and position are saved sequentially. Quantum system helps in representing images include the ability to speed up processing of the image like feature extraction, image segmentation. Merging QIP and quantum machine learning to likely resolve problems which may be

demanding to classical systems, specifically for those which system which may need processing vast volumes of images in various domains such as medical screening, geographic information, and restoration of image.

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